

P510
PHYSICS
Paper 1
2 Hours



CLEVERLAND HIGH SCHOOL-MAYA
MID OF TERM II ASSESSMENT 2025
SENIOR FIVE
PHYSICS
PAPER ONE

INSTRUCTIONS TO LEARNERS;

Attempt all items.

Assume where necessary, acceleration due to gravity. $g = 9.81ms^{-2}$

ITEM 1

- a) A group of aspiring astrophysicists are studying the properties of a new discovered exoplanet, they hypothesize that the orbit period T around the star depends on the following factors;

- i) The gravitational constant, G with dimensions $L^2M^{-1}T^{-2}$*
- ii) The mass, m of the star.*
- iii) The average radius, r of the exoplanet's orbit.*

They want to determine the relationship between these quantities using dimensional analysis.

Task.

Determine the relationship between the orbital period, T , the gravitational constant, G , mass m and the radius.

- b) The student now wants to determine the dimensions of the planet's angular momentum, $\mathbf{N} = \mathbf{I} \mathbf{w}$, where $\mathbf{I}$ is the moment of inertia with dimensions ML^2 and $\mathbf{w}$ is the angular velocity with dimensions T^{-1} .

Task

Determine the dimensions of angular momentum, $\mathbf{N}$

ITEM 2

- a) A model rocket is launched vertically upwards from rest. It accelerates at a constant rate of 5ms^{-2} for 6 seconds, after which the engine cuts off. Neglecting air resistance.

Task

Determine;

- (i) The maximum height reached by the rocket.
 - (ii) The total time the rocket is in the air.
- b) A cliff diver jumps from a 30m high and falls vertically downwards into the sea below.

Task

- (i) Determine how long is the diver in the air.
- (ii) Find the diver's velocity just before hitting the water.

ITEM 3

A builder of mass 80kg is constructing a decorative cone shaped structured using an irregular steel plate. Before rolling the plate into a cone, the builder tests its balance and stability by finding its center of gravity and analyzing the resultant of forces on it.

After obtaining the center of gravity, the builder then tests for a translational equilibrium of the plate by it in a force field of three coplanar forces whose lines of action intersect at the center of gravity of the plate. These forces include the following;

- Force, $F_1 = 15\text{N}$, acting horizontally to the right.
- Force, $F_2 = 20\text{N}$, acting vertically downwards.
- Force, $F_3 = 10\text{N}$, acting at angle of 30° above the positive horizontal.

The builder observed that translational equilibrium is not achieved in this case but does not know the magnitude and direction of the fourth coplanar force that can be added to the existing force field in order to achieve translation equilibrium.

Task

As a learner of physics,

- a) Describe how the builder can determine the center of gravity of plate (steel) using a suspension method and plumb line.
- b) Determine the magnitude and direction of the fourth force for translational equilibrium of the irregular steel plate to be achieved.
- c) Briefly discuss the stability of the decorative cone shaped structure when it rests on its wide base versus when it is balanced on its narrow apex/tip.

END